

Ethics for AI CO4 - Assignment

2310080043-K. Yash Chowdary

- 1. Explain how different countries regulate AI (national and global approaches).

National level:

- European Union: Focuses on strict regulation (risk-based laws like the AI Act).
- United States: Uses guidelines and sector-specific rules rather than one single law.
- China: Strong government control with strict monitoring and data regulations.

- Global level:

- United Nations and OECD promote shared AI ethics principles.

Key idea: No single global law yet—countries follow different models.

- 2. Explain the link between ethical AI and sustainable development.

Ethical AI supports sustainable development by:

- Reducing inequality (fair systems)
- Improving healthcare and education access
- Supporting environmental protection (e.g., climate prediction)

It aligns with the United Nations Sustainable Development Goals by promoting fairness, inclusivity, and long-term benefits.

- 3. Explain ethical problems related to AI leadership and control.

- Concentration of power in big tech companies
- Lack of accountability for AI decisions
- Risk of misuse (surveillance, manipulation)

4. Describe how ethical decisions are made by humans and AI systems.

- **Humans:**
 - Use values, emotions, experience, and moral reasoning
- **AI systems:**
 - Use algorithms, data, and predefined rules

Difference: Humans can understand context and ethics deeply; AI cannot truly “understand” morality.

5. Discuss whether governments are ready for advanced AI technologies.

- **Partially ready:**
 - Some countries have policies and research funding
- **Challenges:**
 - Lack of technical expertise
 - Rapid AI development outpacing laws
 - Limited global coordination

Conclusion: Governments are still catching up.

6. Describe key ethical principles used in AI policy-making with examples.

- **Fairness:** Avoid discrimination (e.g., unbiased hiring AI)
- **Transparency:** Explain decisions (e.g., loan approval AI)
- **Accountability:** Responsibility for outcomes
- **Privacy:** Protect personal data
- **Safety:** Ensure reliable systems

7. Explain the steps involved in ethical decision-making in AI.

- Identify the problem
- Collect relevant data
- Analyze risks and impacts
- Consider ethical principles
- Choose the best solution
- Implement and monitor outcomes
-

8. Suggest ways to improve ethical decision-making in AI systems.

- **Use diverse datasets**
- **Include human oversight**
- **Regular audits and testing**
- **Interdisciplinary teams (ethics + tech)**

- Clear ethical guidelines and training

G. Explain challenges in implementing AI regulations and how to address them.

Challenges:

- Rapid technological change
- Lack of global standards
- Enforcement difficulties
- High cost of compliance

Solutions:

- International cooperation
- Flexible and adaptive laws
- Public-private partnerships
- Continuous monitoring and updates

10. Justify why ethics is important in AI policy-making.

Challenges:

- Rapid technological change
- Lack of global standards
- Enforcement difficulties
- High cost of compliance

Solutions:

- International cooperation
- Flexible and adaptive laws
- Public-private partnerships
- Continuous monitoring and updates